

Pyrolysis of Mixed Plastics in a Fluidized Bed of Hard Burnt Lime

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ABSTRACT: A mixture consisting of 45 wt % polyethylene, 20 wt % polypropylene, 20 wt % polystyrene, and 15 wt % poly(ethylene terephthalate) was pyrolyzed in the presence of steam as well as nitrogen, using a fluidized bed reactor with hard burnt lime (HBL) as bed material. The experiments were carried out at 600 and 700 °C. Unlike soft burnt lime, HBL exhibited good fluidizing properties, with negligible attrition. The impact of HBL on the product distribution and the ability of HBL to support the degradation of PET under certain conditions were investigated. Compared with experiments done in the presence of quartz sand, the gas yield increased, while the wax fraction was significantly reduced. In the presence of HBL, the benzene yield rose sharply due to the decarboxylation of PET. While terephthalic acid was present in the wax fraction derived from experiments with quartz sand, it was not detected as a part of the wax fraction when HBL was used.

INTRODUCTION

Municipal waste plastics (MWP) can be considered a valuable source of fuel since they consist mainly of hydrocarbons, such as polyethylene (PE), polypropylene (PP), and polystyrene (PS). The rising price of crude oil due to increasing demand in developing countries (e.g., China, India) makes it necessary to recover the resources used in making plastics.

The degradation of the six main compounds of MWP, high-density PE (HDPE), low-density PE (LDPE), PP, PS, poly(ethylene terephthalate) (PET), and poly(vinyl chloride) (PVC),¹ has been well investigated. While research has been carried out on the various reactors, including batch reactors,^{2–6} semibatch reactors,^{7,8} fixed bed reactors,⁹ cyclized sphere reactors,¹⁰ autoclaves,^{11–13} tube reactors,¹⁴ and combinations of different reactors,¹⁵ the main focus has been on the degradation behavior of mixed plastics using a fluidized bed reactor.^{16–21}

The presence of catalytic materials has been shown to have a strong impact on the product composition. In fluidized bed reactors, the most common bed material is quartz sand,^{16–21} which, because it is considered inert, is used as a standard for comparison with other bed materials. Studies have also been done on the use of various materials, such as zeolites,^{8,22} red mud,⁷ and aluminum-based catalysts,⁶ in various reactors.

The pyrolysis of plastic mixtures results in gaseous and liquid products, containing mainly aliphatic and aromatic hydrocarbons and some oxygen-containing products from the pyrolysis of PET.^{1,19} De Marco et al.¹¹ observed the formation of char during the pyrolysis of mixed plastics at 500 °C and attributed it to the presence of PET. It was also stated that the presence of PET caused problems during the execution of the experiment due to the evolution of a white material (probably terephthalic acid and benzoic acid), which precipitated on the walls and tubes of the reactor installation. Similar observations were made by Sakata et al.² during the pyrolysis of PE/PET mixtures at 430 °C.

Several attempts have been made to prevent the formation of solids from PET. Obuchi et al.⁹ successfully reduced the amount of solids derived from a PP/PET mixture by using a TiO₂/SiO₂ catalyst. At the same time it was observed that the fraction of high boiling compounds increased with the PET content in the input

material. Masuda et al.¹⁵ degraded PET containing plastics at 500 °C in the presence of steam atmosphere and FeOOH as a catalyst. The waxy product obtained in the absence of the catalyst was converted into oil and gas when the catalyst was used. Yoshioka et al.^{23–26} used Ca(OH)₂ and CaO for the decarboxylation of terephthalic acid derived from the hydrolysis of PET and obtained high yields of benzene. CaO acts as a basic catalyst which forms calcium terephthalate with free terephthalic acid.²⁷ Calcium terephthalate is then decarboxylated, releasing benzene and CO₂ as products. However, Ruppert et al.^{28,29} reported that the Lewis acid sites present in CaO might also have an impact on the product distribution of other plastics.

Even though sometimes CaO is added to the feed of fluidized bed reactors in order to remove halogens,²¹ it is too soft to have been considered as a bed material in fluidized bed reactors. In this study, we investigated the pyrolysis of mixed plastics in a fluidized bed when hard burnt lime (HBL) is used. In general, the catalytic activity of HBL is considered to be low due to the closed surface structure of the sintered material; however, the significant improvement in hardness due to sintering makes it possible to use HBL as a bed material.

EXPERIMENTAL SECTION

Laboratory Scale Pyrolysis Plant. Eight experiments were carried out using a laboratory scale pyrolysis plant with a capacity of 100 g h^{−1} (Figure 1). The fluidized bed reactor used had a free diameter of 51 mm and was loaded with 500 g of bed material, as described in detail by Greve.³⁰ The reactor was electrically heated by two 1000 W half-shells. An addition to the original setup was made by including a steam generator, consisting of a curled copper tube with a length of 3000 mm and a diameter of 10 mm, and a diameter of the loops of 40 mm, placed in a 600 W heater. Plastic material was fed directly into the fluidized bed by a double

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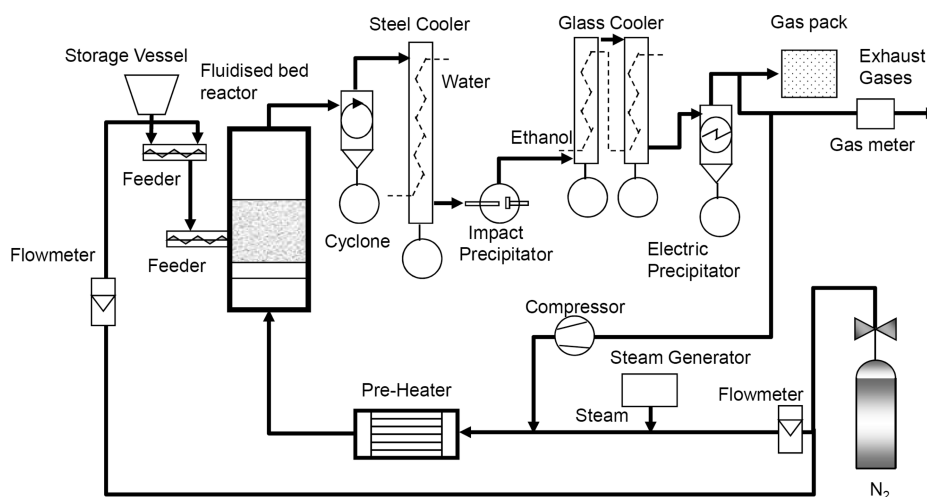


Figure 1. Experimental setup.

conveyor system. The first conveyor was used to control the feeding speed, and the second one was used to quickly transport the material into the reactor. After pyrolysis, the products left the reactor and passed a cyclone for the removal of solid materials. Liquid products were condensed in a steel cooler and two glass coolers. For the experiments at 600 °C, an additional impact precipitator was installed between the steel cooler and the glass coolers in order to remove the waxes from the gas stream. The remaining aerosol was separated by an electric precipitator before the noncondensable gases passed a gas meter and were released into a draft. A gas sample was taken every 30 min.

Feed Material. The plastics used were all delivered by Aldrich. After grinding to a particle size below 0.5 mm, PE (45 wt %), PP (20 wt %), PS (20 wt %), and PET (15 wt %) were mixed in accordance with the municipal waste composition of Sapporo City, but acknowledging only these four main compounds.

Bed Material. Two different bed materials were used for these experiments. The aim of this study was the investigation of CaO as bed material on the degradation of mixed plastics. The first attempts on soft burnt CaO failed since the strong forces between the particles caused the formation of channels. However, it was possible to maintain a working fluidized bed with hard burnt CaO, which was burnt at 1190 °C for 10 min. Blocks of CaO several centimeters across were ground for a short time and sieved. Even after very short grinding periods of 1 or 2 s, about 50% of the material was obtained as fine dust and only 20–30% was obtained in the required particle size range between 300 and 500 μm . The BET surface area and pore volume of the HBL used were determined to be 0.345 $\text{m}^2 \text{g}^{-1}$ and $1.57 \times 10^{-3} \text{cm}^3 \text{g}^{-1}$. Due to the effort required in the preparation of the bed material, the CaO was burnt in a muffle furnace at 950 °C for 6 h in order to regenerate the material. Quartz sand (SiO_2) of the same size was used for reference experiments. The pyrolysis of plastics in fluidized bed reactors using a SiO_2 bed material has been well investigated. Therefore, experiments with SiO_2 were conducted to evaluate the performance of this reactor and for comparisons with the new bed material.

The carbonation and calcination reactions were experimentally studied in a thermogravimetric analyzer (TGA). A 10 mg sample of HBL was heated at 10 K min^{-1} from 50 to 1000 °C with a helium gas flow of 200 mL min^{-1} and kept at that temperature for 30 min. After calcination was completed, the

sample was cooled down, using a cooling rate of 10 K min^{-1} from 1000 to 50 °C with a CO_2 gas flow of 200 mL min^{-1} , and then kept for another 30 min at 50 °C. Using this method, $X_{\text{CaCO}_3} = 0.87 \text{ mol \%}$ of the CaO was converted into CaCO_3 . The penetration depth of CO_2 d_{CO_2} was calculated by

$$d_{\text{CO}_2} = \frac{X_{\text{CaCO}_3}}{\rho_{\text{CaO}}} A_{\text{CaO}}$$

with the CaO density $\rho_{\text{CaO}} = 3.34 \text{ g cm}^{-3}$ and the BET surface area $A_{\text{CaO}} = 0.345 \text{ m}^2 \text{g}^{-1}$. The resulting penetration depth of 0.90 nm shows that only the surface was converted into CaCO_3 , implying that the particle core was inert.

Experimental Conditions. The experiments were carried out at 600 and 700 °C in a nitrogen atmosphere and a steam/nitrogen (2/1) atmosphere. The conditions are given in Table 1.

Analysis. After each experiment, gas, oil, and solid products were obtained. Hydrogen was analyzed by gas chromatography with a thermal conductivity detector (GC-TCD), using a GL Science GC 323 with a Chrompack Carboxplot P7 packed column. Other gases were analyzed by gas chromatography with a flame ionization detector (GC-FID), using a Shimadzu GC-17A with a Chrompack CP Porabond Q capillary column, connected with a GL Science MT 221 methanizer for the determination of carbon oxides. The difference between the organic input and the solid and liquid products was assumed to be the gas mass.

The wax fraction from the impact precipitator and the liquid products from the coolers and the electric precipitator were unified and distilled, resulting in a water phase, oil, and waxes (boiling point > 290 °C). The distillation equipment consisted of a 250 mL round-bottom flask heated by a heating mantle and connected to a 13 cm Vigreux column (inner diameter 13 mm). A thermometer was attached at the connection between the Vigreux column and the water-cooled condenser. The pressure inside the apparatus was controlled by an Eyela NVC-2100 pressure controller (Tokyo, Japan) connected with a Vacuum-brand MD-1C membrane vacuum pump (Wertheim, Germany). The condensed oil was frozen in a flask cooled with liquid nitrogen in order to avoid losses at low pressure. The product oil was heated slowly until the temperature at the head of the Vigreux column reached 100 °C. Then the pressure was reduced

Table 1. Conditions of Experiments

	experiment							
	1	2	3	4	5	6	7	8
temperature [°C]	600	600	600	600	700	700	700	700
bed material	sand	sand	HBL	HBL	sand	sand	HBL	HBL
fluidizing gas	N ₂	steam	N ₂	steam	N ₂	steam	N ₂	steam
input [g]	160.8	128.7	178.9	199.5	199.4	188.1	198.3	200.3
feeding time [h]	2.78	3.00	2.85	2.85	2.57	2.30	3.00	1.60
feeding rate [g h ⁻¹]	57.8	42.9	62.8	70.0	77.7	81.8	66.1	125.2
temperature before fluidized bed [°C]	450.4	471.5	365.0	383.5	501.0	541.3	444.6	400.9
temperature of fluidized bed [°C]	599.2	599.2	596.0	597.0	698.7	699.2	699.0	697.7
temperature of freeboard [°C]	567.0	570.8	586.0	596.0	656.2	666.8	662.4	664.9
N ₂ flow [L/min]	7.5	2.5	7.5	2.5	7.5	2.5	7.5	2.5
steam flow [L/min]	0	5	0	5	0	5	0	5
residence time [s]	2.2	2.2	2.2	2.2	2.0	2.0	2.0	2.0
pressure before fluidized bed [kPa]	7.7	5.9	8.5	7.1	8.1	7.2	6.8	4.3
pressure of freeboard [kPa]	3.9	2.0	5.1	3.9	4.0	3.1	3.2	1.3

slowly to 2.0 kPa, while the temperature was kept constant. After that, the temperature was allowed to rise to 160 °C, corresponding to the fluorene cut (297 °C at 101.5 kPa).

The oil and water phases were analyzed by GC–MS, using an HP6890 GC coupled with an HP5973 mass spectrometer, and quantified by GC-FID, using a GL Science GC 390. For both GC–MS and GC-FID, the same column was used (GL Science Inert Cap SMS/Sil). The high boiling residue was burnt for 6 h at 815 °C in order to determine the content of the inorganic material.

The bed material and cyclone fraction were also burnt for 6 h at 815 °C in order to determine their organic derived contents. The results are given as organic residue in Table 2. For the X-ray diffraction (XRD) investigation, a Rigaku RINT 2200 VHF+/PC was used.

Waxes were analyzed by Fourier transformed infrared spectroscopy (FT-IR) using a Bio-Rad Win IR-165.

RESULTS AND DISCUSSION

SiO₂ Experiments. The experiments with SiO₂ as a bed material were carried out at two different temperatures and under two different atmospheres: nitrogen and steam (Table 2). Since the effect of the atmosphere was negligible, it will be referred to only when the differences are worth noting.

At 600 °C in a nitrogen atmosphere, low boiling oil and waxes were obtained as the main products with about 45 and 36 wt %, respectively; gases yielded 18 wt %. The amount of organic residue remained below 2 wt %. The oil consisted mainly of aliphatic compounds between C₅ and C₁₇. High yields of styrene were obtained due to the decomposition of PS. Other aromatics deriving from the degradation of PS were ethylbenzene and α -methylstyrene. The styrene content significantly increased in the steam atmosphere, and reached 13 wt % after 7.5 wt % in nitrogen atmosphere. More than half of the produced gases were butenes (10 wt %). All other gases remained below 1 wt % with the exception of propene (1.3 wt %).

By raising the temperature to 700 °C, the gas production in the nitrogen atmosphere increased by a factor of 3, reaching 51 wt %, the same as has been observed by other researchers.^{1,17} At this

higher temperature, the oil fraction (37 wt %) and wax fraction (9.8 wt %) were reduced, while the organic residue slightly increased. It should be noted that in the presence of steam not only was the wax fraction reduced but less organic residue was observed. Less oil was obtained at the higher temperature with a significant change in the oil composition. Aliphatic compounds, which constituted a large part of the oil at 600 °C, underwent aromatization at 700 °C and various benzene and naphthalene derivatives were formed. The main product at this temperature was still styrene (9.7 wt %); however, this was followed by benzene (5.2 wt %), which was almost absent at 600 °C. Aliphatic compounds, mainly alkenes and dienes, were limited to carbon numbers between C₅ and C₈.

The fractions of most of the gases increased, with the ethene and propene rising in the nitrogen atmosphere from 1 to 13 wt % and from 2 to 13 wt %, respectively, due to the enhanced depolymerization of PE and PP. On the other hand, butenes increased only slightly when the temperature rose from 600 to 700 °C. The amounts of carbon oxides increased from 0.86 to 2.7 wt % mainly due to the enhanced degradation of PET at the higher temperature. The use of steam resulted in a further rise of several products. The fractions of both ethene and propene increased slightly to 16 and 15 wt %, respectively, and the butene fraction rose from 13 wt % in nitrogen to 16 wt % in a steam atmosphere. The enhanced production of carbon dioxide in the steam atmosphere was probably due to the more efficient decarboxylation of PET. In the presence of steam and when the temperature is sufficiently high enough, PET is hydrolyzed first and the resulting terephthalic acid is decarboxylated. If the ester linkage is not hydrolyzed, the first step in PET degradation is the random fission of the polymer chain, leaving a carboxyl and a vinyl end group (Scheme 1).³¹ In the end, oxygen-containing compounds such as acetophenone are formed. This was supported by the fact that in a steam atmosphere less oxygen-containing products were released. In total, the use of steam increased the gas yield at 700 °C from 51 to 60 wt %. The gas fraction obtained by Kaminsky and Kim¹⁹ at about 700 °C was about 10 wt % less due to the use of pyrolysis gas as a fluidizing gas, which resulted in a higher yield of aromatics, while aliphatics were almost absent in the liquid fraction.

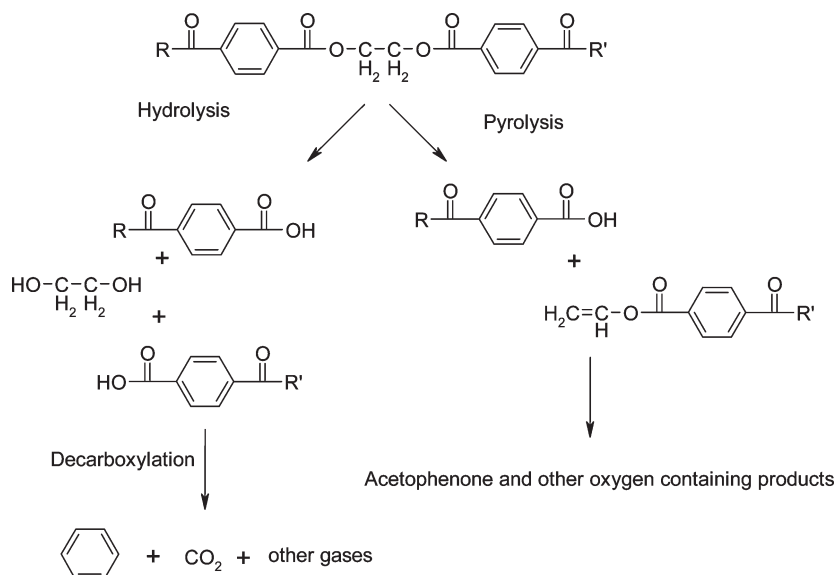
Table 2. Mass Balance of the Pyrolysis of Mixed Plastics

		experiment							
		1	2	3	4	5	6	7	8
temperature [°C]		600	600	600	600	700	700	700	700
bed material		sand	sand	HBL	HBL	sand	sand	HBL	HBL
fluidizing gas		N ₂	steam	N ₂	steam	N ₂	steam	N ₂	steam
gas [wt %]	gas total	18	17	41	43	51	60	62	65
	hydrogen	0.07	0.14	0.73	0.33	0.18	0.18	0.84	0.12
	carbon monoxide	—	1.05	2.0	0.81	1.3	1.2	1.1	6.0
	carbon dioxide	0.86	—	5.0	10	2.7	4.0	4.7	15
	methane	0.32	0.80	2.0	2.0	3.1	3.9	4.2	3.9
	ethylene	0.69	1.7	7.8	8.3	13	16	17	15
	ethane	0.95	0.46	1.5	1.7	2.1	2.1	2.7	1.6
	propylene	1.3	2.7	9.4	10	13	15	16	12
	propane	0.25	0.38	0.51	0.71	0.57	0.61	0.68	0.22
	C ₄	10	9.3	8.1	9.2	13	16	14	10
	unknown	2.9	0.50	3.6	0.19	1.6	1.2	2.1	1.0
oil [wt %]	oil total	45	48	45	49	37	34	31	31
	aromatic total	10	16	12	21	20	17	20	22
	benzene	0.01	0.05	3.6	5.2	5.2	5.3	7.0	5.2
	toluene	0.17	0.29	0.25	0.93	2.1	1.7	1.1	1.7
	styrene	7.5	13	6.1	12	9.7	7.4	8.2	11
	ethylbenzene	0.59	1.00	0.57	0.86	0.42	0.41	0.52	0.79
	xylene	—	—	—	0.18	0.08	0.07	0.10	0.17
	1- <i>H</i> -indene	—	—	—	0.15	0.33	0.21	0.37	0.26
	α-methylstyrene	0.82	1.1	1.1	1.2	0.70	0.49	0.79	0.78
	C ₃ -benzene	0.53	0.65	0.17	0.52	0.61	0.43	0.57	0.99
	C ₄ -benzene	0.21	0.19	—	0.17	0.23	0.14	0.23	0.18
	C ₅ -benzene	0.12	0.12	—	—	—	—	—	—
	1-methylnaphthalene	—	—	—	—	0.18	0.09	0.17	0.06
	naphthalene	0.32	0.16	0.14	0.12	0.43	0.26	0.58	0.35
	methylnaphthalene	—	—	0.07	—	0.04	0.03	0.14	0.36
	biphenyl	—	—	0.13	—	0.03	0.08	0.32	0.04
	aliphatic total	32	30	29	21	15	16	8.6	6.7
	C ₅	12	11	8.3	7.1	7.8	9.9	6.0	5.0
	C ₆	13	11	5.8	5.0	5.0	4.3	2.1	1.3
	C ₇	3.9	2.5	4.5	2.6	1.6	0.99	0.18	0.20
	C ₈	0.13	0.19	4.5	1.9	0.76	0.53	0.29	0.10
	C ₉	0.62	0.84	0.58	0.77	—	—	—	—
	C ₁₀	0.81	0.86	0.88	0.97	0.09	0.07	0.04	0.17
	C ₁₁	0.64	0.73	0.85	0.61	0.08	0.06	0.04	—
	C ₁₂	0.42	0.55	0.63	0.52	0.06	0.07	—	—
	C ₁₃	0.39	0.80	0.54	0.46	—	—	—	—
	C ₁₄	0.25	0.59	0.80	0.64	—	—	—	—
	C ₁₅	0.16	0.37	0.67	0.42	—	—	—	—
	C ₁₆	0.09	0.24	0.44	0.28	—	—	—	—
	C ₁₇	0.03	0.07	0.49	0.22	—	—	—	—
	oxygen-containing compounds	0.25	0.18	1.7	0.96	0.55	0.25	0.75	0.07
	not identified	2.3	2.0	2.7	5.8	1.3	1.1	1.7	2.9
wax [wt %]		36	34	10	5.4	9.8	4.2	3.6	1.8
organic residue [wt %]		1.8	1.2	3.9	2.3	2.3	1.5	3.0	1.8
total [wt %]		100	100	100	100	100	100	100	100

CaO Experiments. In the presence of CaO, the reduction of waxes (10 wt %) was enhanced in the nitrogen atmosphere at

600 °C and, due to the improved degradation of PET, a rise in gas production (41 wt %) was noted, while the oil fraction remained

Scheme 1



Scheme 2

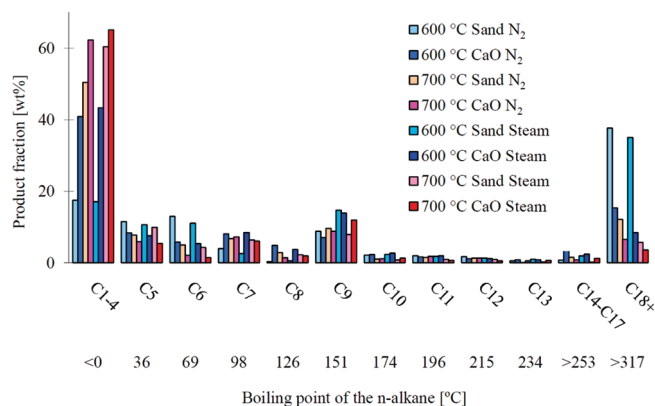
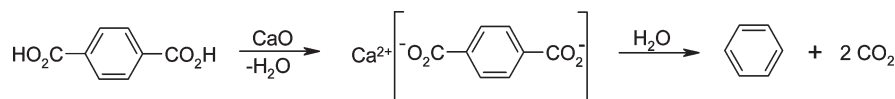


Figure 2. C-NP graph of the pyrolysis oils from the degradation of mixed plastics in the presence of sand and CaO, respectively.

constant at 45 wt %. It should be noted also that, in the presence of CaO, steam had just a small impact on the product distribution.

Even if the oil yield was barely influenced by the addition of CaO, the composition of the oil changed in the presence of CaO. The benzene fraction increased from 0.01 wt % in the absence of CaO to 3.6 wt %. In a steam atmosphere, a benzene fraction of 5.2 wt % was achieved. Since aromatization was still negligible at 600 °C, the aromatic compounds must have originated from PS and PET. The origin of most of the benzene may have been from PET, since PS depolymerizes under the formation of styrene, styrene dimers, and trimers. Thus, the increasing benzene

fraction indicated the increasing hydrolysis of PET and the subsequent decarboxylation of the resulting terephthalic acid. Kumagai et al.²⁷ showed that CaO forms with terephthalic acid calcium terephthalate, which is subsequently decarboxylated. Benzene and carbon dioxide were released (Scheme 2). Since PET promotes the formation of residue during pyrolysis, the effect is also clearly visible in the reduction of the amount of high boiling compounds. Therefore, the amount of products with carbon numbers between 7 and 17 increased from 8 wt % in the absence of CaO to 15 wt % in its presence, regardless of the atmosphere.

The effect of CaO can be seen from the C-NP graph (C-NP, average carbon number) shown in Figure 2, where the product distribution is plotted against the boiling point. This diagram makes use of the fact that hydrocarbons on a nonpolar GC column are roughly separated in order of their boiling points. Aromatic compounds shift to a carbon number increased by 1 due to the slightly different behaviors on the GC column caused by their aromatic character. Therefore, it can be assumed that certain retention times represent certain boiling points. From the C-NP graph in Figure 2, it can be seen that at 600 °C more gases were obtained in the presence of CaO and the fractions of C₇–C₈ and C₁₄–C₁₇ increased. While the rise in the C₇ fraction resulted from the promoted benzene formation due to the decarboxylation of terephthalic acid, the other fractions benefited from the cracking of longer aliphatic chains, causing a reduction in the C₅, C₆ fraction and the high boiling C₁₈+ fraction. The same but weaker tendencies were observed at 700 °C. Like common cracking catalysts, CaO has also catalytic active Lewis acid sites.^{28,29} However, due to the small surface area

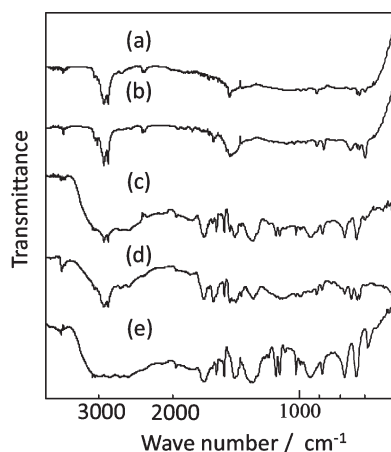


Figure 3. FT-IR spectra of waxes obtained at 600 °C in the presence of (a) HBL and N₂, (b) HBL and steam, (c) sand and N₂, and (d) sand and steam. For comparison purposes, the FT-IR spectrum of TPA (e) is added.

and the basic character of the material, the effect is much weaker than for other cracking catalysts. Even if at 600 °C the C₁₈₊ fraction was reduced in nitrogen and steam by 59 and 76%, the effect was much smaller at 700 °C (nitrogen, 46%; steam, 31%). This suggests that pyrolysis proceeded faster than the cracking reaction. Common cracking catalysts are efficient in a temperature range between 350 and 500 °C.^{32–34} We did not try temperatures lower than 600 °C since we believed the amount of wax produced would have obstructed the separation system. The inability to reduce the temperature clearly shows the limitations of HBL as a cracking catalyst.

The gas yield increased in the presence of CaO mainly due to an increased formation of ethane, propene, and carbon oxides. Carbon oxides yielded only about 1 wt % at 600 °C in the absence of CaO, indicating the insufficient degradation of PET. The degradation of PET in the presence of CaO, however, resulted in 5.0 and 10 wt % carbon dioxide in the nitrogen and steam atmospheres, respectively. The rise in the benzene yield and the reduction in the amount of high boiling products can also be seen as indicators of the improved degradation of PET. The increased production of ethene and propene also indicates that CaO has properties of a cracking catalyst.

Very high hydrogen yields were observed in a nitrogen atmosphere. The amount of hydrogen produced at 600 and 700 °C was 0.73 and 0.84 wt %, respectively. This observation correlated with an increased production of carbonaceous residue. These high hydrogen yields were not observed in the presence of sand as bed material nor when steam and CaO were used. This indicates that CaO catalyzed the formation of char. However, steam seems to have the ability to inhibit the char formation by deactivating the active sites responsible for the formation of the residue. Since the cracking efficiency of CaO was even higher in the steam atmosphere, it can be assumed that the sites responsible for the formation of char and for the polymer cracking were not the same.

The efficiency of PET decarboxylation is also visible in the FT-IR spectra of the pyrolysis waxes (Figure 3). Due to the high sublimation point of about 400 °C, terephthalic acid remains after the distillation in the wax fraction. The FT-IR spectrum of terephthalic acid (TPA) was characterized by the C=O

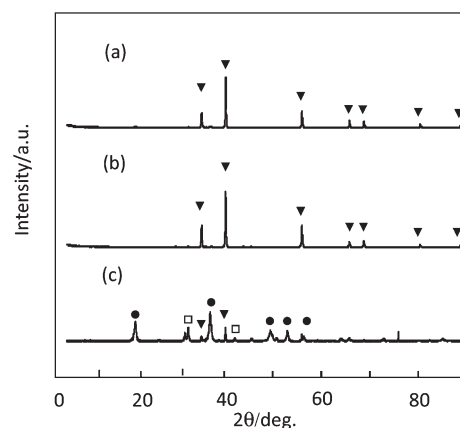


Figure 4. XRD patterns of HBL: (a) bed material before reaction, (b) bed material after reaction, and (c) cyclone fraction, each obtained at 600 °C in steam atmosphere. (▼) CaO; (●) Ca(OH)₂; (□) CaCO₃.

stretching vibration at 1686 cm⁻¹ and the C–O stretching vibration at 1284 cm⁻¹ (Figure 3e). The aromatic C–C stretching vibrations of terephthalic acid were also visible at 1521 and 1579 cm⁻¹. All these vibrations were also observed when quartz sand was used as the bed material (Figure 3c,d). The experiments in the presence of HBL, however, resulted in waxes, showing neither C=O nor C–O vibrations (Figure 3a,b), suggesting the absence of any organic acids as well as the absence of other carbonyl compounds. While aliphatic C–H vibrations at 2885 and 2950 cm⁻¹ were observed in all the waxes, the waxes obtained in the presence of CaO showed especially strong aromatic C–H vibrations at 3030 cm⁻¹ and alkyne C–H vibrations at 2330 cm⁻¹.

The effect of the reaction on HBL can be seen at the cyclone fraction. In contrast to the bed material that remained in the reactor throughout the whole experiment, from the initial heating period over the reaction time and the final cooling process, the cyclone fraction was obtained during the experiment, undergoing a rapid cooling step that left the material in a state close to that during the reaction. Furthermore, besides the leftover fine dust from the grinding process, the cyclone fraction consisted solely of material from the attrition of the bed material, allowing for an inside look into the properties of the bed material. While the bed material before and after the reaction consisted solely of CaO (Figure 4a,b), Ca(OH)₂ and CaCO₃ were also present in the cyclone fraction, suggesting that active catalytic sites were present at the surface of the HBL. Even though the core of the HBL might have been inactive, these surface sites were capable of reducing the TPA concentration below the detection limit at a reaction temperature of 600 °C. Ca(OH)₂ and CaCO₃ might have been formed during the short residence time in the cyclone at a temperature of about 300 °C. However, the presence of Ca(OH)₂ in the reactor cannot be completely excluded, since it can be observed under certain conditions even at temperatures higher than 600 °C.³⁵

CONCLUSION

Using HBL as a bed material and steam as fluidizing gas can reduce the yields of waxes by decarboxylating TPA, which results in an increase in the gas yields. When HBL is used as a bed material at 600 °C, the yields of useful products, such as oils and

gases, was comparable to those at 700 °C when quartz sand was used. This indicates that HBL is an efficient catalyst for the feedstock recycling of mixed plastics by pyrolysis. It was also confirmed that the hardness of HBL makes it suitable for use as a bed material in a fluidized bed reactor under the conditions employed. The bed material was reused in several experiments without a reduction in performance.

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REFERENCES

- (1) Williams, E. A.; Williams, P. T. The Pyrolysis of Individual Plastics and a Plastic Mixture in a Fixed Bed Reactor. *J. Chem. Technol. Biotechnol.* **1997**, 70, 9–20.
- (2) Sakata, Y.; Uddin, M. A.; Koizumi, K.; Murata, K. Thermal Degradation of Polyethylene Mixed with Poly(vinyl chloride) and Poly(ethyleneterephthalate). *Polym. Degrad. Stab.* **1996**, 53, 111–117.
- (3) Bhaskar, T.; Matsui, T.; Kaneko, J.; Uddin, M. A.; Muto, A.; Sakata, Y. Novel Calcium Based Sorbent (Ca-C) for the Dehalogenation (Br, Cl) Process During Halogenated Mixed Plastic (PP/PE/PS/PVC and HIPS-Br) Pyrolysis. *Green Chem.* **2002**, 4, 372–375.
- (4) Bhaskar, T.; Uddin, M. A.; Murai, K.; Kaneko, J.; Hamano, K.; Kusaba, T.; Muto, A.; Sakata, Y. Comparison of Thermal Degradation Products from Real Municipal Waste Plastic and Model Mixed Plastics. *J. Anal. Appl. Pyrolysis* **2003**, 70, 579–587.
- (5) Bhaskar, T.; Kaneko, J.; Muto, A.; Sakata, Y.; Jakab, E.; Matsui, T.; Uddin, M. A. Pyrolysis Studies of PP/PE/PS/PVC and HIPS-Br Plastics Mixed with PET and Dehalogenation (Br, Cl) of the Liquid Products. *J. Anal. Appl. Pyrolysis* **2004**, 72, 27–33.
- (6) Zhou, Q.; Tang, C.; Wang, Y.-Z.; Zheng, L. Catalytic Degradation and Dechlorination of PVC-Containing Mixed Plastics Via Al-Mg Composite Oxide Catalysts. *Fuel* **2004**, 83, 1727–1732.
- (7) Yanik, J.; Uddin, M. A.; Sakata, Y. The Effect of Red Mud on the Liquefaction of Waste Plastics in Heavy Vacuum Gas Oil. *Energy Fuels* **2000**, 15, 163–169.
- (8) Kim, J.-R.; Van, J.-H.; Park, D.-W.; Lee, M.-H. Catalytic Degradation of Mixed Plastics Using Natural Clinoptilolite Catalyst. *React. Kinet. Catal. Lett.* **2004**, 81, 73–81.
- (9) Obuchi, E.; Suyama, M.; Nakano, K. Decomposition of Mixed Plastics Consisting of Polypropylene and Polyethylene Terephthalate into Oils over Titania/Silica Catalysts. *J. Mater. Cycles Waste Manage.* **2001**, 3, 88–92.
- (10) Karayıldırım, T.; Yanik, J.; Yüksel, M.; Sağlam, M.; Haussmann, M. Degradation of PVC Containing Mixtures in the Presence of HCl Fixators. *J. Polym. Environ.* **2005**, 13, 365–374.
- (11) de Marco, I.; Caballero, B.; Torres, A.; Laresgoiti, M. F.; Chomón, M. J.; Cabrero, M. A. Recycling Polymeric Wastes by Means of Pyrolysis. *J. Chem. Technol. Biotechnol.* **2002**, 77, 817–824.
- (12) Siddiqui, M. N. Conversion of Hazardous Plastic Wastes into Useful Chemical Products. *J. Hazard. Mater.* **2009**, 167, 728–735.
- (13) Siddiqui, M. N.; Redhwi, H. H. Pyrolysis of Mixed Plastics for the Recovery of Useful Products. *Fuel Process. Technol.* **2009**, 90, 545–552.
- (14) Miskolczi, N.; Bartha, L.; Angyal, A. Pyrolysis of Polyvinyl Chloride (PVC)-Containing Mixed Plastic Wastes for Recovery of Hydrocarbons. *Energy Fuels* **2009**, 23, 2743–2749.
- (15) Masuda, T.; Kushino, T.; Matsuda, T.; Mukai, S. R.; Hashimoto, K.; Yoshida, S. Chemical Recycling of Mixture of Waste Plastics Using a New Reactor System with Stirred Heat Medium Particles in Steam Atmosphere. *Chem. Eng. J.* **2001**, 82, 173–181.
- (16) Kaminsky, W.; Schlesselmann, B.; Simon, C. M. Thermal Degradation of Mixed Plastic Waste to Aromatics and Gas. *Polym. Degrad. Stab.* **1996**, 53, 189–197.
- (17) Kim, J.-S.; Kaminsky, W.; Schlesselmann, B. Pyrolysis of a Fraction of Mixed Plastic Wastes Depleted in PVC. *J. Anal. Appl. Pyrolysis* **1997**, 40–41, 365–372.
- (18) Williams, E. A.; Williams, P. T. Analysis of Products Derived from the Fast Pyrolysis of Plastic Waste. *J. Anal. Appl. Pyrolysis* **1997**, 40–41, 347–363.
- (19) Kaminsky, W.; Kim, J.-S. Pyrolysis of Mixed Plastics into Aromatics. *J. Anal. Appl. Pyrolysis* **1999**, 51, 127–134.
- (20) Kaminsky, W.; Schmidt, H.; Simon, C. M. Recycling of Mixed Plastics by Pyrolysis in a Fluidised Bed. *Macromol. Symp.* **2000**, 152, 191–199.
- (21) Cho, M.-H.; Jung, S.-H.; Kim, J.-S. Pyrolysis of Mixed Plastic Wastes for the Recovery of Benzene, Toluene, and Xylene (Btx) Aromatics in a Fluidized Bed and Chlorine Removal by Applying Various Additives. *Energy Fuels* **2009**, 24, 1389–1395.
- (22) Halász, J.; Kónya, Z.; Faragó, Z. T.; Sievert, K.; Kiricsi, I. Degradation of Pure and Waste Polyolefins and PVC in the Presence of Modified Porous Catalysts. *Stud. Surf. Sci. Catal.* **2008**, 174 (Part 2), 1021–1026.
- (23) Yoshioka, T.; Kitagawa, E.; Mizoguchi, T.; Okuwaki, A. High Selective Conversion of Poly(ethylene terephthalate) into Oil Using Ca(OH)₂. *Chem. Lett.* **2004**, 33, 282–283.
- (24) Yoshioka, T.; Handa, T.; Grause, G.; Lei, Z.; Inomata, H.; Mizoguchi, T. Effects of Metal Oxides on the Pyrolysis of Poly(ethylene terephthalate). *J. Anal. Appl. Pyrolysis* **2005**, 73, 139–144.
- (25) Yoshioka, T.; Grause, G.; Otani, S.; Okuwaki, A. Selective Production of Benzene and Naphthalene from Poly(butylene terephthalate) and Poly(ethylene naphthalene-2,6-dicarboxylate) by Pyrolysis in the Presence of Calcium Hydroxide. *Polym. Degrad. Stab.* **2006**, 91, 1002–1009.
- (26) Grause, G.; Handa, T.; Kameda, T.; Mizoguchi, T.; Yoshioka, T. Effect of Temperature Management on the Hydrolytic Degradation of PET in a Calcium Oxide Filled Tube Reactor. *Chem. Eng. J.* **2011**, 166, 523–528.
- (27) Kumagai, S.; Grause, G.; Kameda, T.; Takano, T.; Horiuchi, H.; Yoshioka, T. Decomposition of Gaseous Terephthalic Acid in the Presence of CaO. *Ind. Eng. Chem. Res.* **2011**, 50, 1831–1836.
- (28) Ruppert, A. M.; Meeldijk, J. D.; Kuipers, B. W. M.; Erné, B. H.; Weckhuysen, B. M. Glycerol Etherification over Highly Active CaO-Based Materials: New Mechanistic Aspects and Related Colloidal Particle Formation. *Chem.—Eur. J.* **2008**, 14, 2016–2024.
- (29) Calatayud, M.; Ruppert, A. M.; Weckhuysen, B. M. Theoretical Study on the Role of Surface Basicity and Lewis Acidity on the Etherification of Glycerol over Alkaline Earth Metal Oxides. *Chem.—Eur. J.* **2009**, 15, 10864–10870.
- (30) Greve, S. Chemical Recycling of Soft PVC Products in the Fluidized Bed Pyrolysis and Completed Investigations on the Behavior of Flame-Protected Plastics with GC/MS. Dissertation, University of Hamburg, Hamburg, 2003.
- (31) Montaudo, G.; Puglisi, C.; Samperi, F. Primary Thermal Degradation Mechanisms of PET and PBT. *Polym. Degrad. Stab.* **1993**, 42, 13–28.
- (32) Olazar, M.; Lopez, G.; Amutio, M.; Elordi, G.; Aguado, R.; Bilbao, J. Influence of FCC Catalyst Steaming on HDPE Pyrolysis Product Distribution. *J. Anal. Appl. Pyrolysis* **2009**, 85, 359–365.
- (33) Marcilla, A.; Hernández, M. d. R.; García, A. N. Degradation of LDPE/VGO Mixtures to Fuels Using a FCC Equilibrium Catalyst in a Sand Fluidized Bed Reactor. *Appl. Catal., A* **2008**, 341, 181–191.
- (34) Lin, H.-T.; Huang, M.-S.; Luo, J.-W.; Lin, L.-H.; Lee, C.-M.; Ou, K.-L. Hydrocarbon Fuels Produced by Catalytic Pyrolysis of Hospital

Plastic Wastes in a Fluidizing Cracking Process. *Fuel Process. Technol.* **2010**, *91*, 1355–1363.

(35) Materić, V.; Edwards, S.; Smedley, S. I.; Holt, R. Ca(OH)_2 Superheating as a Low-Attrition Steam Reactivation Method for CaO in Calcium Looping Applications. *Ind. Eng. Chem. Res.* **2010**, *49*, 12429–12434.